

Bent Marshak Waves - Cylindrical 2D Model

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1 Physical Setup and Geometry

We consider radiative heat transport in an optically thick medium inside a cylindrical tube. The longitudinal coordinate is denoted by z , and the transverse coordinate by the radial coordinate r . We assume full axisymmetry:

$$\frac{\partial}{\partial \theta} = 0.$$

The quantity of interest is

$$U(z, r, t) \equiv T^4(z, r, t),$$

which satisfies a diffusion-type equation under the assumptions of **constant opacity** and negligible hydrodynamic motion.

2 Governing Equation

In the steady diffusion region behind the Marshak front, the radiation energy density satisfies Laplace's equation. In axisymmetric cylindrical coordinates this reads:

$$\frac{\partial^2 U}{\partial z^2} + \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial U}{\partial r} \right) = 0. \quad (1)$$

3 Separation of Variables

We seek separable solutions of the form

$$U(z, r) = Z(z)R(r).$$

Substituting into Eq. (1) and dividing by ZR yields

$$\frac{1}{Z} \frac{d^2 Z}{dz^2} + \frac{1}{Rr} \frac{d}{dr} \left(r \frac{dR}{dr} \right) = 0.$$

Setting both terms equal to a separation constant $-\kappa^2$ gives

$$\frac{d^2 Z}{dz^2} - \kappa^2 Z = 0, \quad (2)$$

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{dR}{dr} \right) + \kappa^2 R = 0. \quad (3)$$

4 Radial Eigenfunctions

The radial equation is Bessel's equation of order zero. Its regular solution at $r = 0$ is

$$R(r) = J_0(\kappa r).$$

The general axisymmetric solution therefore takes the form

$$U(z, r) = \sum_{n \geq 0} J_0(\kappa_n r) [A_n(t) e^{\kappa_n z} + B_n(t) e^{-\kappa_n z}]. \quad (4)$$

5 Boundary Conditions

5.1 Axis Regularity

Regularity at the symmetry axis requires

$$\left. \frac{\partial U}{\partial r} \right|_{r=0} = 0.$$

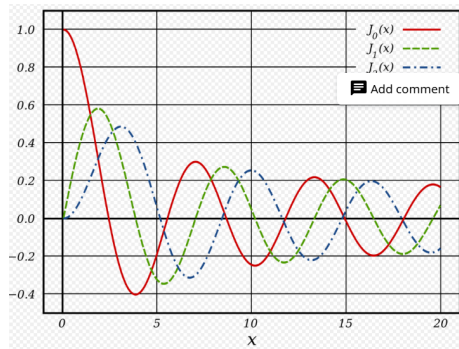


Figure 1: The first three Bessel functions of the first kind, J_0 , J_1 , and J_2 .

For that reason, we want the r -spatial part to be at its peak at $r=0$, and not 0, which is only satisfied by $J_0(\kappa r)$.

5.2 Radiative Wall Losses

At the cylindrical wall $r = R$, radiative energy is lost according to a Marshak-type Robin boundary condition:

$$\left. \frac{\partial U}{\partial r} \right|_{r=R} = \frac{3}{4} \rho \kappa_{opacity} (a - 1) U(R), \quad (5)$$

where a is the wall albedo.

Using the Bessel function identity (Eq. ?? and Eq. ??)

$$\frac{d}{dr} J_0(\kappa r) = -\kappa J_1(\kappa r),$$

the boundary condition yields the eigenvalue equation

$$\boxed{\kappa R J_1(\kappa R) = \varepsilon J_0(\kappa R)}, \quad (6)$$

definition of small parameter:

$$\varepsilon \equiv \frac{3}{4} (1 - a) \tau$$

where $\tau = \rho \kappa_{opacity} R$ is the optical depth.

Solutions are intersections of: $\kappa R J_1(\kappa R) = \varepsilon J_0(\kappa R)$. with small corrections.

This equation determines the discrete spectrum $\{\kappa_n\}$.

Small-loss limit. For $\varepsilon \ll 1$, we can express the the zeroth and first bessel functions using their series representation:

$$J_0(\kappa x) \approx 1 - \frac{(\kappa x)^2}{4}, \quad J_1(\kappa x) \approx \frac{(\kappa x)}{2} - \frac{(\kappa x)^3}{16}.$$

Substituting into Eq. (??) we get:

$$\kappa R \left(\frac{\kappa R}{2} - \frac{(\kappa R)^3}{16} \right) = \varepsilon \left(1 - \frac{(\kappa R)^2}{4} \right).$$

or leading order in κR ,

$$\frac{(\kappa R)^2}{2} \approx \varepsilon, \implies \kappa_0 R \approx \sqrt{2\varepsilon}.$$

6 Marshak Front Representation

The Marshak front position is expanded in the same radial eigenfunctions:

$$z_F(r, t) = \sum_{n \geq 0} c_n(t) J_0(\kappa_n r), \quad (7)$$

where $\{\kappa_n\}$ are determined by the Robin wall condition $\kappa R J_1(\kappa R) = \varepsilon J_0(\kappa R)$.

At leading order, only the lowest mode is retained:

$$z_F(r, t) \approx c_0(t) J_0(\kappa_0 r). \quad (8)$$

Front Evolution Equation: explicit leading-mode reduction

We now derive the ODE for $c_0(t)$ from the front energy balance. Following the Hurricane–Hammer steep-front closure, the front speed is proportional to the longitudinal radiative flux at the front:

$$\rho e \dot{z}_F = -\frac{4\sigma}{3\kappa_R\rho} \left. \frac{\partial U}{\partial z} \right|_{z=z_F}. \quad (9)$$

Here κ_R is the (Rosseland) opacity and σ is the Stefan–Boltzmann constant; e is an effective energy-per-volume parameter for heating the material.

Step 1: leading eigenmode form of $U(z, r, t)$. From the separated solution (Eq. (??)) we retain only the lowest radial eigenfunction:

$$U(z, r, t) \approx J_0(\kappa_0 r) \left(A(t)e^{\kappa_0 z} + B(t)e^{-\kappa_0 z} \right). \quad (10)$$

We impose two longitudinal conditions (as in the planar derivation):

6.1 Source boundary at $z = 0$

From (??) at $z = 0$:

$$T_s^4 = \sum_{n=0}^{\infty} (A_n + B_n) J_0(k_n r). \quad (11)$$

Using orthogonality of $J_0(k_n r)$ on $[0, R]$ with weight r , this implies that to leading order only the fundamental mode carries the constant function:

$$A_0 + B_0 = T_s^4 + O(\epsilon), \quad A_{n \geq 1} + B_{n \geq 1} = O(\epsilon). \quad (12)$$

(Exactly as in the planar derivation: the constant-in- r boundary drives primarily the lowest mode.)

6.2 Front boundary condition: weak-curvature implementation (full derivation)

The bent Marshak front is the moving surface

$$z = z_F(r, t),$$

at which the temperature drops sharply. In the steep-front approximation we impose

$$U(z_F(r, t), r, t) \approx 0.$$

Because the wall losses are assumed weak (high albedo), the curvature of the front is small. We therefore expand the front position about a mean (or center-line) position $c_0(t)$:

$$z_F(r, t) = c_0(t) + \delta z_F(r, t), \quad \delta z_F = O(\epsilon). \quad (13)$$

Taylor expansion of the front boundary condition. Expand $U(z, r, t)$ about $z = c_0(t)$:

$$\begin{aligned} U(z_F(r, t), r, t) &= U(c_0 + \delta z_F, r, t) \\ &= U(c_0, r, t) + \delta z_F(r, t) \partial_z U(c_0, r, t) + \frac{1}{2} \delta z_F(r, t)^2 \partial_z^2 U(c_0, r, t) + \cdots. \end{aligned} \quad (14)$$

Imposing $U(z_F, r, t) \approx 0$ and retaining only the leading term in the weak-curvature limit gives

$$\boxed{U(c_0, r, t) = 0 \quad (\text{leading order in curvature}).} \quad (15)$$

Higher-order corrections can be incorporated by keeping the $\delta z_F \partial_z U$ term; here we focus on the leading-order closure.

modal form of the solution in the diffusion region. In the quasi-steady diffusion region behind the front, U satisfies Laplace's equation and admits the separated expansion

$$U(z, r, t) = \sum_{n=0}^{\infty} J_0(k_n r) \left(A_n(t) e^{k_n z} + B_n(t) e^{-k_n z} \right). \quad (16)$$

impose the leading-order front condition mode-by-mode. Insert $z = c_0$ into (??) and enforce (??):

$$0 = U(c_0, r, t) = \sum_{n=0}^{\infty} J_0(k_n r) \left(A_n e^{k_n c_0} + B_n e^{-k_n c_0} \right).$$

Since the set $\{J_0(k_n r)\}$ forms an orthogonal basis on $0 \leq r \leq R$ (with weight r), the coefficient of each mode must vanish:

$$\boxed{A_n(t) e^{k_n c_0(t)} + B_n(t) e^{-k_n c_0(t)} = 0 \quad \forall n.} \quad (17)$$

This immediately gives

$$B_n(t) = -A_n(t) e^{2k_n c_0(t)}. \quad (18)$$

rewrite the longitudinal dependence in hyperbolic-sine form. Start with the generic z -dependence of mode n :

$$A_n e^{k_n z} + B_n e^{-k_n z}.$$

Substitute (??):

$$\begin{aligned} A_n e^{k_n z} + B_n e^{-k_n z} &= A_n e^{k_n z} - A_n e^{2k_n c_0} e^{-k_n z} \\ &= A_n \left(e^{k_n z} - e^{k_n (2c_0 - z)} \right) \\ &= A_n e^{k_n c_0} \left(e^{k_n (z - c_0)} - e^{-k_n (z - c_0)} \right) \\ &= 2A_n e^{k_n c_0} \sinh(k_n (z - c_0)). \end{aligned} \quad (19)$$

It is convenient to factor out a normalization $\sinh(k_n c_0)$ so that the remaining coefficient is dimensionless and (later) directly tied to the source boundary condition. Define a new amplitude $\tilde{A}_n(t)$ by

$$\tilde{A}_n(t) \equiv 2A_n(t)e^{k_n c_0(t)} \sinh(k_n c_0(t)). \quad (20)$$

Then (??) becomes the compact “sinh-form”

$$\boxed{A_n e^{k_n z} + B_n e^{-k_n z} = \tilde{A}_n(t) \frac{\sinh(k_n(z - c_0(t)))}{\sinh(k_n c_0(t))}.} \quad (21)$$

This expression automatically satisfies the front condition because at $z = c_0$ the numerator vanishes.

insert back into the full field expansion. Substituting (??) into (??) yields

$$U(z, r, t) = \sum_{n=0}^{\infty} J_0(k_n r) \tilde{A}_n(t) \frac{\sinh(k_n(z - c_0(t)))}{\sinh(k_n c_0(t))}. \quad (22)$$

normalize by the source temperature and define dimensionless amplitudes. Let T_s be the imposed source temperature (for constant drive) so that the natural scale is T_s^4 . Define dimensionless coefficients

$$\alpha_n(t) \equiv -\frac{\tilde{A}_n(t)}{T_s^4}. \quad (23)$$

Then (??) becomes

$$\boxed{\frac{U(z, r, t)}{T_s^4} = -\sum_{n=0}^{\infty} \alpha_n(t) J_0(k_n r) \frac{\sinh(k_n(z - c_0(t)))}{\sinh(k_n c_0(t))}.} \quad (24)$$

This is exactly the cylindrical analogue of the paper’s planar expression (their Eq. (27)), with $\cos(\cdot)$ replaced by $J_0(\cdot)$ and transverse eigenvalues set by the Robin condition.

how α_n are fixed by the source boundary condition. At $z = 0$ the boundary condition is $U(0, r, t) = T_s^4$ (constant in r for a uniform drive). Plugging $z = 0$ into (??) gives

$$1 = \frac{U(0, r, t)}{T_s^4} = -\sum_{n=0}^{\infty} \alpha_n J_0(k_n r) \frac{\sinh(-k_n c_0)}{\sinh(k_n c_0)} = \sum_{n=0}^{\infty} \alpha_n J_0(k_n r),$$

since $\sinh(-x) = -\sinh(x)$. Therefore

$$\boxed{1 = \sum_{n=0}^{\infty} \alpha_n J_0(k_n r) \quad (0 \leq r \leq R),} \quad (25)$$

i.e. the constant function 1 is expanded in the $J_0(k_n r)$ basis. This uniquely determines the α_n via Bessel orthogonality (with weight r). In the high-albedo limit, the lowest mode is nearly uniform, so

$$\alpha_0 = 1, \quad \alpha_{n \geq 1} = O(\epsilon),$$

matching the weak-loss ordering used in the perturbation theory. Therefore, to leading order we can set $\alpha_0 = 1$ and $\alpha_{n \geq 1} = 0$, giving the leading-mode approximation

$$\boxed{\frac{U(z, r, t)}{T_s^4} \approx -\alpha_0 J_0(k_0 r) \frac{\sinh(k_0(z - c_0(t)))}{\sinh(k_0 c_0(t))} = J_0(k_0 r) \frac{\sinh(k_0(c_0(t) - z))}{\sinh(k_0 c_0(t))}.}$$
(26)

Step 2: compute the front gradient $\partial_z U|_{z=z_F}$. Differentiate (??) with respect to z :

$$\begin{aligned} \partial_z U(z, r, t) &\approx U_s(t) J_0(\kappa_0 r) \frac{\partial}{\partial z} \left[\frac{\sinh(\kappa_0(c_0 - z))}{\sinh(\kappa_0 c_0)} \right] = \\ &= -U_s(t) J_0(\kappa_0 r) \frac{\kappa_0 \cosh(\kappa_0(c_0 - z))}{\sinh(\kappa_0 c_0)}. \end{aligned}$$

Evaluating at the front $z = c_0(t)$ gives

$$\partial_z U|_{z=c_0(t)} \approx -U_s(t) J_0(\kappa_0 r) \frac{\kappa_0}{\sinh(\kappa_0 c_0(t))}. \quad (27)$$

Step 3: project the front balance onto the lowest eigenmode. Equation (??) is a *pointwise* statement along the front. In the leading-mode truncation we enforce it in a weighted (Galerkin) sense by multiplying both sides by $J_0(\kappa_0 r)$ and integrating over the cross section with cylindrical weight $r dr$:

$$\int_0^R \rho e \dot{z}_F(r, t) J_0(\kappa_0 r) r dr = -\frac{4\sigma}{3\kappa_R \rho} \int_0^R \partial_z U|_{z=z_F} J_0(\kappa_0 r) r dr. \quad (28)$$

Using $z_F(r, t) \approx c_0(t) J_0(\kappa_0 r)$, the left-hand side becomes

$$\int_0^R \rho e \dot{c}_0(t) J_0^2(\kappa_0 r) r dr = \rho e \dot{c}_0(t) \mathcal{N}_0, \quad \mathcal{N}_0 \equiv \int_0^R J_0^2(\kappa_0 r) r dr.$$

Using (??), the right-hand side becomes

$$-\frac{4\sigma}{3\kappa_R \rho} \int_0^R \left(-U_s(t) J_0(\kappa_0 r) \frac{\kappa_0}{\sinh(\kappa_0 c_0)} \right) J_0(\kappa_0 r) r dr = \frac{4\sigma}{3\kappa_R \rho} \frac{U_s(t) \kappa_0}{\sinh(\kappa_0 c_0)} \mathcal{N}_0.$$

Canceling the common factor $\mathcal{N}_0$ yields the closed ODE

$$e \dot{c}_0(t) = \frac{4\sigma}{3\kappa_R \rho^2} \frac{U_s(t) \kappa_0}{\sinh(\kappa_0 c_0(t))}. \quad (29)$$

Step 4: define an effective diffusion constant D_M . For a constant drive $U_s(t) = U_0$ (or for slow variation treated quasi-statically), we define the effective constant

$$D_M \equiv \frac{8\sigma U_0}{3e \kappa_R \rho^2}, \quad (30)$$

so that (??) becomes

$$\dot{c}_0(t) = \frac{D_M \kappa_0}{2} \frac{1}{\sinh(\kappa_0 c_0(t))}. \quad (31)$$

Solution for the Isotropic Mode: integration to $\cosh^{-1}$

Multiply (??) by $\kappa_0 \sinh(\kappa_0 c_0)$:

$$\kappa_0 \sinh(\kappa_0 c_0) \dot{c}_0 = \frac{D_M \kappa_0^2}{2}.$$

Recognizing the left-hand side as a time derivative,

$$\frac{d}{dt} \cosh(\kappa_0 c_0) = \kappa_0 \sinh(\kappa_0 c_0) \dot{c}_0,$$

we obtain

$$\frac{d}{dt} \cosh(\kappa_0 c_0) = \frac{D_M \kappa_0^2}{2}. \quad (32)$$

Integrating from $t = 0$ with $c_0(0) = 0$ gives

$$\cosh(\kappa_0 c_0(t)) = 1 + \frac{D_M \kappa_0^2 t}{2}.$$

Therefore,

$$\boxed{c_0(t) = \frac{1}{\kappa_0} \cosh^{-1} \left(1 + \frac{D_M \kappa_0^2 t}{2} \right)}. \quad (33)$$

Finally, the leading-order front shape is

$$\boxed{z_F(r, t) \approx c_0(t) J_0(\kappa_0 r) = \frac{1}{\kappa_0} \cosh^{-1} \left(1 + \frac{D_M \kappa_0^2 t}{2} \right) J_0(\kappa_0 r)}. \quad (34)$$

One can make the derivation even better if defining $D = D_M(1 + \frac{\varepsilon}{2})$ (higher order accuracy):

$$\boxed{z_F(r, t) \approx c_0(t) J_0(\kappa_0 r) = \frac{1}{\kappa_0} \cosh^{-1} \left(1 + \frac{D \kappa_0^2 t}{2} \right) J_0(\kappa_0 r)}. \quad (35)$$

Time-dependent drive $U_s(t)$ (non-constant boundary temperature)

So far, the closed-form solution (??) was obtained assuming a *constant* drive $U_s(t) = U_0$ (equivalently, constant boundary temperature T_s so that $U_s = T_s^4$). We now generalize the derivation to an *arbitrary* time-dependent drive $U_s(t)$.

Where $U_s(t)$ enters. The only place where the drive appears in the leading-mode reduction is through the boundary condition at $z = 0$,

$$U(0, r, t) = U_s(t),$$

which implies $A(t) + B(t) = U_s(t)$ (Eq. (??)). Repeating the same algebra as in (??) (with $U_0 \rightarrow U_s(t)$), we obtain the front gradient

$$\partial_z U|_{z=c_0(t)} \approx -U_s(t) J_0(\kappa_0 r) \frac{\kappa_0}{\sinh(\kappa_0 c_0(t))}. \quad (36)$$

Therefore, the Galerkin projection step (Eq. (??)) yields the same ODE structure as before, but with an explicit $U_s(t)$ factor:

$$\dot{c}_0(t) = \frac{\kappa_0}{\sinh(\kappa_0 c_0(t))} \frac{4\sigma}{3e \kappa_R \rho^2} U_s(t). \quad (37)$$

Define a time-dependent diffusion coefficient. It is convenient to define

$$D_M(t) \equiv \frac{8\sigma}{3e \kappa_R \rho^2} U_s(t), \quad (38)$$

so that Eq. (??) becomes

$$\dot{c}_0(t) = \frac{D_M(t) \kappa_0}{2} \frac{1}{\sinh(\kappa_0 c_0(t))}. \quad (39)$$

Integration for general $U_s(t)$. Multiply Eq. (??) by $\kappa_0 \sinh(\kappa_0 c_0)$ to obtain

$$\frac{d}{dt} \cosh(\kappa_0 c_0(t)) = \frac{\kappa_0^2}{2} D_M(t).$$

Integrating from 0 to t with $c_0(0) = 0$ gives

$$\boxed{\cosh(\kappa_0 c_0(t)) = 1 + \frac{\kappa_0^2}{2} \int_0^t D_M(t') dt' = 1 + \frac{4\sigma \kappa_0^2}{3e \kappa_R \rho^2} \int_0^t U_s(t') dt'.} \quad (40)$$

Hence,

$$\boxed{c_0(t) = \frac{1}{\kappa_0} \cosh^{-1} \left(1 + \frac{\kappa_0^2}{2} \int_0^t D_M(t') dt' \right) = \frac{1}{\kappa_0} \cosh^{-1} \left(1 + \frac{4\sigma \kappa_0^2}{3e \kappa_R \rho^2} \int_0^t U_s(t') dt' \right).} \quad (41)$$

Finally, the leading-order cylindrical front shape becomes

$$\boxed{z_F(r, t) \approx c_0(t) J_0(\kappa_0 r) = \frac{J_0(\kappa_0 r)}{\kappa_0} \cosh^{-1} \left(1 + \frac{4\sigma \kappa_0^2}{3e \kappa_R \rho^2} \int_0^t U_s(t') dt' \right).} \quad (42)$$

Special case: power-law drive. If $U_s(t) = U_0 \left(\frac{t}{t_0}\right)^p$ for $t \geq 0$, then

$$\int_0^t U_s(t') dt' = \frac{U_0}{p+1} \frac{t^{p+1}}{t_0^p},$$

and Eq. (??) becomes

$$c_0(t) = \frac{1}{\kappa_0} \cosh^{-1} \left(1 + \frac{4\sigma \kappa_0^2}{3e \kappa_R \rho^2} \cdot \frac{U_0}{p+1} \frac{t^{p+1}}{t_0^p} \right).$$

This reduces to the constant-drive expression when $p = 0$.

7 Physical Interpretation

- The transverse structure is governed by $J_0(\kappa_0 r)$ rather than $\cos(k_0 y)$.
- Radiative wall losses introduce curvature and retard the front.
- Cylindrical geometry enhances edge cooling relative to planar geometry.
- The leading-order dynamics are entirely controlled by the lowest eigenvalue κ_0 .

The correspondence between planar and cylindrical formulations is:

$$\cos(ky) \longleftrightarrow J_0(\kappa r), \quad \tan(kL) \longleftrightarrow \frac{\kappa R J_1(\kappa R)}{J_0(\kappa R)}.$$

All higher-order perturbation theory carries over with this substitution.

8 Useful bessel function identities

Series Representation:

$$J_n(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma(m+n+1)} \left(\frac{x}{2}\right)^{2m+n} \quad (43)$$

Properties:

$$2nJ_n(x) = x(J_{n-1}(x) + J_{n+1}(x)) \quad (44)$$

$$J_n(-x) = (-1)^n J_n(x), \quad J_{-n} = (-1)^n J_n(x) \quad (45)$$

Differentiation,

$$\frac{d}{dx} J_n(x) = \frac{1}{2} (J_{n-1}(x) - J_{n+1}(x)) \quad (46)$$

$$\frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x), \quad \frac{d}{dx} [x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x) \quad (47)$$

Asymptotic form for large and small x :

$$J_n(x) \approx \frac{1}{n!} \left(\frac{x}{2}\right)^n, \quad x \ll 1 \quad (48)$$

$$J_n(x) \approx \sqrt{\frac{2}{\pi x}} \cos\left(x - \frac{n\pi}{2} - \frac{\pi}{4}\right), \quad x \gg 1 \quad (49)$$

And lastly the orthogonality relation:

$$\int_0^R r J_n\left(\frac{\alpha_m r}{R}\right) J_n\left(\frac{\alpha_k r}{R}\right) dr = \frac{R^2}{2} [J_{n+1}(\alpha_m)]^2 \delta_{mk} \quad (50)$$

where α_m is the m 'th root of $J_n(x)$.

Albedo Derivation

Consider a gold cylindrical shell surrounding foam.

Let

$$F_{\text{in}} = \sigma_{SB} T_s^4 \quad (51)$$

$$F_{\text{out}} = \sigma_{SB} T_{\text{obs}}^4 \quad (52)$$

The detector measures the re-emitted flux:

$$\sigma_{SB} T_{\text{obs}}^4 = \sigma_{SB} T_s^4 + \frac{F(0, t)}{2} \quad (53)$$

Thus,

$$\sigma_{SB} T_s^4 - \frac{F(0, t)}{2} = \sigma_{SB} T_{\text{obs}}^4 \quad (54)$$

Since

$$F(0, t) = \dot{E}_w \quad (55)$$

we define the albedo:

$$a(t) = \frac{\text{reflected flux}}{\text{incident flux}} = \frac{F_{\text{in}}}{F_{\text{out}}} = \frac{\sigma_{SB} T_0^4 + \frac{\dot{E}_w}{2}}{\sigma_{SB} T_0^4 - \frac{\dot{E}_w}{2}} \quad (56)$$